

HSUAN YA (MICHAEL) TSAI

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PROFESSIONAL SUMMARY

Senior infrastructure and networking engineer with 14+ years of experience building and operating high-performance networking systems across large-scale cloud and security platforms. Experienced in FPGA-based network offload, Mellanox NIC integration, GPU infrastructure, and production-scale Azure networking supporting millions of servers. Strong background in C/C++, Linux, live-site operations, and end-to-end ownership of networking features from platform onboarding through production deployment.

TECHNICAL EXPERTISE

Networking: Azure Accelerated Networking, SR-IOV, FPGA offload, Mellanox NICs, VNet Encryption, TCP/IP, DPI

Languages and tools: C, C++, C#, Python, PowerShell, Perl, Linux, Jenkins, Google Test, lex/yacc

PROFESSIONAL EXPERIENCE

MICROSOFT | Azure Core Compute / Host Networking

Redmond, WA

Senior Software Engineer

Sep. 2021 - Present

Software Engineer II

Apr. 2019 - Aug. 2021

- Developed and evolved Azure Accelerated Networking, an FPGA-based network-offload platform that moves packet processing from software into hardware, contributing to its evolution from approximately 30 Gbps to 400 Gbps networking.
- Owned Mellanox NIC integration on the Azure side, including driver qualification, validation, rollout, telemetry analysis, and production issue mitigation across millions of Azure servers.
- Expanded Azure Accelerated Networking to specialized A100 and H100 GPU SKUs used for OpenAI workloads, enabling hardware-accelerated networking and diagnosing platform-specific issues during onboarding and production rollout.
- Enabled Accelerated Networking and VNet Encryption across new Azure hardware, VM, and OS platforms, including new VM generations and Azure Boost SKUs.
- Built host-networking servicing and infrastructure components in C++, along with automated validation and qualification tooling in Python and PowerShell.
- Served in extensive on-call and live-site rotations, diagnosing networking failures, analyzing telemetry, mitigating production incidents, and deploying fixes in large-scale Azure production environments; also mentored junior engineers.

EARLIER EXPERIENCE

MICROSOFT | Windows Protocols

Shanghai, China

Software Engineer II

Nov. 2017 - Apr. 2019

- Developed protocol test suites in C# to validate interoperability and networking behavior across Windows releases, with continuous integration using Jenkins.

DELL INC.

Taipei, Taiwan

Senior Software Engineer

Aug. 2017 - Oct. 2017

- Developed software components for Alienware gaming PCs using C/C++ and C#.

TREND MICRO INC. | Taipei, Taiwan

Project Lead / Senior Software Engineer

Jan. 2015 - Aug. 2017

Software Engineer

Apr. 2012 - Jan. 2015

- Led a 5-person network engine team developing network security and advanced-threat detection capabilities in C/Linux.
- Built Deep Packet Inspection functionality across L2-L4 and application protocols, including TCP, UDP, ICMP, IP, and HTTP.
- Developed real-time threat-detection DSL capabilities and improved engineering infrastructure, test automation, and unit-test tooling using C/C++, Python, Perl, lex/yacc, and Google Test.

EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA

Los Angeles, CA

Master of Science in Computer Science (General)

Sep. 2010 - Dec. 2011

NATIONAL CHIAO TUNG UNIVERSITY

Hsinchu, Taiwan

Master of Science in Computer Science

Sep. 2007 - Jul. 2009

Thesis: The Design of a P2P Live and Time-Shift Streaming System

NATIONAL CHIAO TUNG UNIVERSITY

Hsinchu, Taiwan

Bachelor of Science in Computer Science

Sep. 2002 - Jun. 2007

GPA: 3.70 (26/207)